

On the Meaning of Temperature and Pressure in Statistical Mechanics

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Abstract

The understanding of concept often depends on the standpoint and perspective from which we approach it. In thermodynamics, temperature and pressure are understood through molecular kinetic theory; in statistical mechanics, however, they can be understood differently. This paper attempts to clarify the statistical meaning of temperature and pressure from the perspective of statistical mechanics. It also discusses new interpretations of several familiar macroscopic physical phenomena. These interpretations appear more concise and direct than those given from a dynamical perspective.

Introductory Remarks

In general, the macroscopic state of a system is commonly characterized by three parameters: temperature, pressure, and volume. We are very familiar with the concepts of temperature and pressure in thermodynamics, and we have also come to understand them anew through certain relations statistical mechanics. However, the position and meaning of temperature and pressure in statistical mechanics have not yet been fully clarified. This paper will elucidate this point through a thermal-equilibrium radiation system.

Furthermore, by exploring this meaning in greater depth, we will find that statistical mechanics has its own distinctive perspective for approaching problems—analyzing them in terms of the overall structure of a system. This is entirely different from the way traditional dynamical theories—from the perspective of individual entities—approach problems. From its distinctive perspective, statistical mechanics provides a new understanding of the intrinsic nature of things. Taking temperature and pressure as examples, this paper examines these two macroscopic parameters entirely from a statistical perspective, and the resulting understanding will be markedly different from that obtained from a dynamical perspective.

Among the three commonly used macroscopic parameters, volume is a physical quantity associated with spatial intuition, whereas temperature and pressure are physical quantities that reflect the microscopic state of a system. It can therefore be considered that “temperature” and “pressure” establish a direct connection between microscopic structure and macroscopic phenomena. Furthermore, we will find that, in statistical physics, the two concepts of temperature and pressure do not need to be defined independently; rather, they are manifested through the relationship between the microscopic and macroscopic levels. Likewise, their position and meaning in statistical mechanics are revealed through this relationship.

In my previous paper, I discussed the statistical meaning of temperature, and the present paper will further develop this study. In addition, this paper will present the statistical meaning

of pressure in statistical physics. It will also provide a qualitative explanation of how pressure varies with temperature in a thermal-equilibrium radiation system. This paper further discusses, from the perspective of statistical mechanics, new interpretations of several familiar macroscopic physical phenomena; these new interpretations appear more concise and direct than descriptions from a dynamical perspective.

I. Theoretical Basis

The theoretical framework and statistical derivation have already been presented in the previous paper [1, 2, 3]. To ensure the overall completeness of the present paper, some important steps from the theoretical basis are give again below.

The phase space of a system can be divided into phase cells of size h^3 . As proposed by Bose [4], the microscopic distribution of energy among the phase cells in a blackbody radiation system determines the macroscopic properties of the radiation system.

A radiation system contains phase cells of different energies. The energy of the s -th phase cell is denoted by E^s (s ranging from 0 to ∞). In a blackbody radiation system, $E^s = h\nu^s$ (ν is the frequency of the quantum). The volume of any phase cell can be expressed as $dx dy dz dp_x dp_y dp_z$. For given the spatial range and momentum range, the number of phase cells in the system is

$$\Gamma_{\text{tot}} = \frac{1}{h^3} \int dx dy dz dp_x dp_y dp_z. \quad (1)$$

For a given total energy E , using the energy-momentum relation for a quantum, $E = pc$, and taking into account the two polarization directions, the total number of phase cells in the system at energy E , denoted by Γ , is

$$\Gamma = \frac{8\pi V}{3h^3} \left(\frac{E}{c}\right)^3. \quad (2)$$

Now let the number of phase cells within a unit energy interval be Δs , and suppose that Δn quanta are distributed among them, with both Δs and Δn being very large numbers. We use $p_r \Delta s$ to denote the number of phase cells in Δs in which r energy quanta appear. p_r also denotes the probability that r energy quanta appear in the s -th phase cell; therefore, we use p_r^s to replace p_r . Clearly, for all phase cells, the following condition is satisfied,

$$\sum_r p_r^s = 1. \quad (3)$$

For given p_r^s , the number of possible ways of distributing Δn quanta among Δs phase cells is

$$\frac{\Delta s!}{\prod_{r=0}^{\infty} (p_r^s \Delta s)!}. \quad (4)$$

Here, r ranges from 0 to ∞ . Taking the logarithm of the above expression and then using Stirling's formula together with Eq.(3), we obtain

$$-\Delta s \sum_r (p_r^s \ln p_r^s). \quad (5)$$

For the radiation system as a whole, this expression can be written as a sum over all phase cells. Combining it with Boltzmann's principle, one obtains the expression for the entropy S of the radiation system,

$$S = -k_B \sum_{s,r} (p_r^s \ln p_r^s), \quad (6)$$

where k_B is Boltzmann constant.

When the radiation system reaches thermodynamic equilibrium, the entropy is required to attain its maximum value, while the total energy of the system is given by

$$\bar{E} = \sum_{s,r} E^s r p_r^s. \quad (7)$$

Here, $\sum_{s,r} r p_r^s$ is the average number of quanta in the s -th phase cell. We now seek the maximum value of Eq.(6) under the constraint of Eq.(7). Introducing the Lagrange multiplier β , and taking Eq.(3) into account, we obtain

$$p_r^s = (1 - e^{-\beta E^s}) e^{-\beta E^s r}. \quad (8)$$

The expression for β is then determined. Substituting Eq.(8) into Eq.(6), we obtain

$$S = -k_B \sum_s \ln(1 - e^{-\beta E^s}) + k_B \beta \bar{E}. \quad (9)$$

Combining this with the thermodynamic relation $\frac{\partial S}{\partial E} = \frac{1}{T}$, where T denotes the system temperature, gives $\beta = \frac{1}{k_B T}$. Therefore, combining Eq.(7) and Eq.(8), the average energy $\bar{E}^s$ in the s -th phase cell is

$$\bar{E}^s = \frac{E^s}{e^{E^s/k_B T} - 1}. \quad (10)$$

Replacing E^s by $h\nu^s$, we obtain the expression for the average energy of a Planck oscillator.

II. Temperature and Pressure in Statistical Mechanics

2.1 The Statistical Meaning of Temperature

According to the above derivation, under the constraint of the total energy alone, only one Lagrange multiplier is introduced, yielding the expression for the average energy in a phase cell given by Eq.(10). Dividing Eq.(10) by E^s , we obtain a dimensionless mathematical result, which we denote by $\bar{n}^s$. It means the average occupation number of quanta in the s -th phase cell (hereafter referred to simply as the occupation number), and its explicit form is [5]

$$\bar{n}^s = \frac{1}{e^{E^s/k_B T} - 1}. \quad (11)$$

The occupation number indicates the position and meaning of temperature in statistical mechanics. A detailed analysis follows.

According to the statistical theory described above, structure is established as the primary object of study, with the central focus being the distribution of the system's energy among the phase cells. That is, by examining how the microscopic structure composed of phase cells is occupied by quanta, we can describe the microscopic state and macroscopic behavior of the system.

It can be seen that the occupancy number $\bar{n}^s$ is a statistical quantity. Moreover, when the occupancy number changes, the temperature of the system changes as well. This may be understood as follows: the distribution of energy over phase cells and temperature are not in a causal relation, but rather in a relation of representation and what is represented. That is, temperature is a statistical representation of the distribution of the system's energy over phase cells. Put differently, the microscopic distribution of energy within the structure of the system is expressed

macroscopically through the parameter of temperature. In this sense, temperature has a definite statistical meaning.

It follows that “temperature” is derived from the relationship between the microscopic structural distribution of energy and its form of macroscopic manifestation. It does not need to be described through certain entities, as is required from a dynamical perspective. Thus, in statistical mechanics, the concept of “temperature” is understood anew—and it is entirely different from a dynamical perspective.

2.2 The Statistical Meaning of Pressure

The Helmholtz free energy F of the system is defined as $F = \bar{E} - TS$. Substituting the result of Eq.(9), we obtain

$$F = k_B T \sum_s \ln(1 - e^{-\beta E^s}). \quad (12)$$

The thermodynamic differential form of the free energy is: $dF = -pdV - SdT$, and hence $p = -(\frac{\partial F}{\partial V})_T$. Applying Eq.(12) and taking Eq.(2) into account, the pressure of the system is expressed as

$$p = \frac{\bar{E}}{3V} = \frac{1}{3}\bar{u}. \quad (13)$$

Here, $\bar{u} = \bar{E}/V$, which denotes the total energy on the phase cells per unit volume of the system. The above equation can, of course, also be written as

$$\bar{u} = 3p. \quad (14)$$

It can be seen that Eq.(14) expresses the relationship between the microscopic statistical quantity $\bar{u}$ and the macroscopic parameter p . On this basis, we can understand pressure in statistical mechanics as follows: the total energy of the phase cells in the microscopic distribution per unit volume of the system, is understood through the observable macroscopic parameter p —we call it “pressure”. At the same time, pressure establishes a direct connection between the microscopic distribution and macroscopic phenomena, and it is also a statistical representation of the distribution of the system’s energy among the phase cells.

It is worth noting that this interpretation of pressure differs from its definition from a dynamical perspective. On other words, pressure in statistical mechanics does not need to be understood from the perspective of mechanical action. Furthermore, pressure in statistical physics does not need to be defined independently as a concept; rather, it emerges from the relationship between microscopic statistics and macroscopic manifestation.

2.3 A Qualitative Statistical Explanation of the Relationship Between Temperature and Pressure in a Thermal-Equilibrium Radiation System

According to thermodynamic theory, the following relation between the radiation pressure p and the radiation energy density $\bar{u}$ can be obtained: $p = \frac{1}{3}\bar{u}$, and $\bar{u} = aT^4$, where a is a constant [6]. Therefore,

$$p = \frac{1}{3}aT^4. \quad (15)$$

That is, pressure is proportional to the fourth power of temperature. Based on the preceding discussion, the relationship can be explained qualitatively as follows: when the system temperature is manifested as rising (or falling), it means that the total energy of the phase cells per

unit volume must increase (or decrease); macroscopically, this is manifested as an increase (or decrease) in pressure.

III. Some Natural Phenomena

In summary, we can see that statistical mechanics enriches the content of temperature and pressure, and broaden our understanding of them. Below, some new interpretations of natural macroscopic physical phenomena from the perspective of statistical mechanics will be presented.

3.1 Understanding and Description of Thermal Equilibrium and the Zeroth Law of Thermodynamics in Statistical Mechanics

In statistical mechanics, thermal equilibrium is understood as a stable statistical distribution of energy among the phase cells under the constraint that the total energy of the system remains constant. (It should be noted here that this description of thermal equilibrium does not involve dynamical descriptions concerning time, interactions, and so on.) That is, thermal equilibrium is simply the manifestation of a stable distribution of energy among the phase cells, and is recognized macroscopically through temperature.

More generally, if two systems undergo only thermal contact and eventually reach a state of thermal equilibrium, we already know that all systems in thermal equilibrium with one another have the same temperature, regardless of the specific substances within the two systems. In thermodynamics, this is regarded as an empirical result; however, in statistical mechanics, it can be understood microscopically as follows:

If two systems undergo only thermal contact, the total energy of the two systems will be redistributed among the total phase cells. The distribution follows the principle of maximum entropy (that is, the number of microscopic states of the system is maximized), ultimately resulting in a stable distribution structure, which is manifested through the macroscopic parameter of temperature.

Similarly, the zeroth law of thermodynamics can also be understood statistically. In thermodynamics, the zeroth law is defined as follows: If each of two thermodynamic systems A and B is in thermal equilibrium with a third thermodynamic system C, then the two thermodynamic systems A and B must also be in thermal equilibrium with each other. From the statistical microscopic perspective, this can be interpreted as follows [7]: when A, B, and C form a closed composite system, the total energy will be redistributed among the phase cells of the composite system. Following the principle of maximum entropy, a stable distribution temperature value. Therefore, if system C is removed, system A and system B must also be in thermal equilibrium at the same temperature.

3.2 Description of the Phenomenon of Temperature Rise (or Fall) in Statistical Mechanics

If a system undergoes heat transfer with the external environment and subsequently experiences a rise (or fall) in temperature, this process can be described as follows: after the system absorbs (or releases) energy, the total energy is redistributed among the phase cells (the distribution follows the principle of maximum entropy); subsequently, the average number of quanta and the

average energy in each phase cell increase (or decrease), which is macroscopically manifested as a rise (or fall) in the temperature of the system.

It is worth mentioning that, in the above description of the phenomenon of temperature rise (or fall), no concept of an “entity”—for example, a particle—is introduced; there are only phase cells and energy, and the distribution of energy among the phase cells is macroscopically represented by temperature.

3.3 A Non-Dynamical Interpretation of the Pressure of an Ideal Gas

Previously, in Einstein’s paper on the quantum theory of a monatomic ideal gas [1], the form of the energy in a phase cell of a monatomic ideal gas system was already expressed as the kinetic energy E_k , and the traditional relation between kinetic energy and momentum was used: $p^2 = 2mE_k$. The expression for the Helmholtz free energy was then written down. Based on the relation between pressure and the Helmholtz free energy, an expression for the pressure p_{id} of a monatomic ideal gas was obtained

$$p_{id} = \frac{2}{3} \overline{u_k} \quad (16)$$

Here, $\overline{u_k}$ denotes the sum of the kinetic energies in all phase cells per unit volume of the system. Thus, a non-dynamical description of the pressure of a monatomic ideal gas is obtained—it represents the amount of total kinetic energy in the phase cells per unit volume.

In addition, the equation of state for an ideal gas is $pV = Nk_B T$ (where N is the number of molecules in the ideal gas). When the system volume remains unchanged, the relationship between pressure and temperature can also be qualitatively explained in statistical mechanics: when the system temperature is manifested as rising (or falling), the kinetic energy in each phase cell increases (or decreases), and the pressure necessarily increases (or decreases).

IV. Some Supplementary Notes

We can see that, from the statistical perspective, the concept of an “entity” is not introduced, and the object of study in statistical mechanics is always the structure as a whole. It expresses this structure through how energy is distributed among the phase cells. From the point of view, some macroscopic phenomena can be understood and described more directly.

Furthermore, from the perspective of molecular kinetic theory, the microscopic nature of temperature reflects the degree of irregular motion of a large number of molecules within a substance. However, from the perspective of statistical mechanics, temperature reflects the distribution of energy among microscopic phase cells (with different forms of energy). Thus, here temperature does not simply describe the average state of motion of a large number of molecules; rather, it more comprehensively reflects the microscopic energy distribution of the system.

Finally, I would like to remind the reader to reconsider macroscopic thermal phenomena from the microscopic perspective of statistical mechanics. I think that, in addition to the examples given above, there should be others. It is hoped that more examples can be included, enriching the content of statistical mechanics and allowing for a richer understanding of natural phenomena.

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